

Bidirectional EN ↔ ES translation transformer

ENCODER

Reading the diagram

—— data flow

- - - - residual connection

- - - - encoder to decoder

37.6M parameters

4 encoder + 4 decoder layers

d_model 512, 8 heads, d_ff 2048

one shared 16k vocabulary

embeddings tied 3 ways

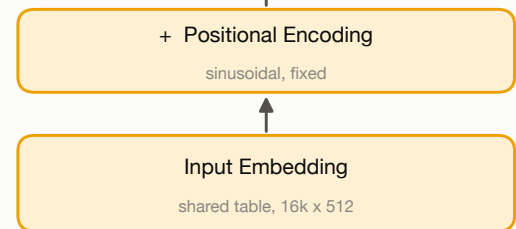
the only path from source to target

memory
(6, 512)

K, V

4x

4x

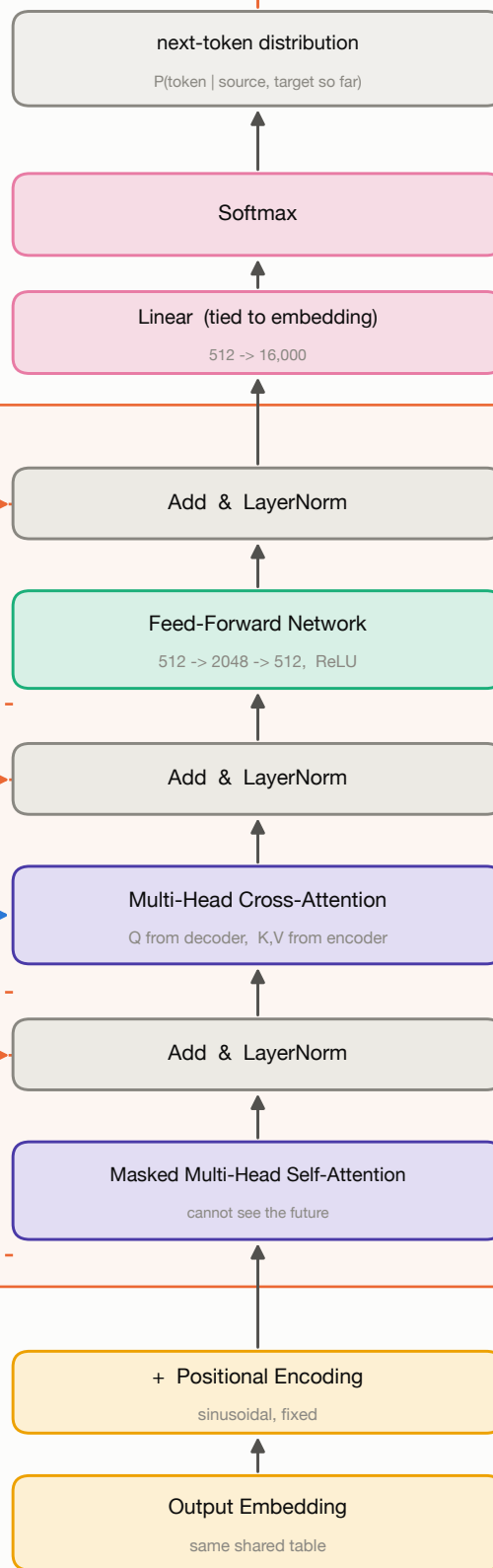


source: <2es> I am tired . </s>

the direction tag tells one model which way to translate

DECODER

Estoy cansado . </s>



target so far: <s> Estoy cansado .

shifted right: position t predicts token t+1